

Table 2: Challenges in serverless computing and their solutions

Challenge	Solution
Cold starts	Use provisioned concurrency or schedule periodic invocations to keep functions warm.
Vendor lock-in	Design for portability by using open standards and adopting a multi-cloud or hybrid strategy.
Limited execution time	Break tasks into smaller functions or use containers and other services for long-running processes.
Resource limitations	Optimise code and delegate heavy processing or storage needs to external services.
Monitoring and debugging	Implement distributed tracing, and use cloud-native monitoring tools (e.g., AWS X-Ray, Azure Monitor).
Security	Enforce strict access controls, implement API gateways, and monitor for vulnerabilities continuously.
Cost management	Optimise function code to minimise execution time, and use cost-monitoring tools to track and manage expenses.
Concurrency limits	Optimise function design for concurrency or request higher limits from the provider if needed.
State management	Leverage external state management services like databases or distributed caches.
Latency sensitivity	Reduce cold start impact, optimise network usage, and deploy functions in regions close to users.

automatically allocate resources based on the number of incoming requests, ensuring optimal performance without over-provisioning.

Cost efficiency: With serverless, users only pay for the actual compute time their code consumes. This pay-per-use model eliminates the need to maintain idle resources, significantly reducing costs, especially for applications with variable workloads.

Increased developer productivity: By removing the burden of infrastructure management, serverless computing allows developers to focus purely on coding and improving the user experience. This leads to:

- Faster development cycles.
- Improved collaboration between development and operations teams, enhancing the DevOps culture.
- Easier integration of microservices, where different parts of an application can

be developed, deployed, and scaled independently.

Optimised resource utilisation:

Serverless architecture ensures that resources are utilised only when needed. Since the infrastructure scales up or down based on real-time usage, it eliminates resource wastage, making cloud environments more energy-efficient and cost-effective. This efficiency not only reduces costs but also aligns with sustainable IT practices by reducing the carbon footprint of data centres.

Support for emerging technologies:

The serverless model is proving to be a strong enabler for emerging technologies such as AI, machine

learning, and IoT. These technologies require massive scalability and on-demand resource availability, which serverless computing provides. By eliminating infrastructure constraints, it allows developers to easily integrate machine learning models, AI services, or IoT data streams into their applications without having to worry about resource management.

In a nutshell, serverless computing is a powerful cloud computing model that offers significant benefits for developers and businesses. By understanding the concepts and use cases, you can determine if serverless is the right choice for your applications. **END** 🐧

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